MILITARY SPECIFICATION

GRENADE, GP, HE, XM74
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament Research and Development Command and is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not normally covered by the drawings and provides Quality Assurance provisions for the fabrication of parts, assembly and packing for one type of grenade designated as Grenade, GP, HE, XM74.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS (See 6.8)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9332580 - Grenade, GP, HE, XM74.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings, specifications, and standards.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing 9332580 and associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Radiographic examination of M219A1E1 grenade fuze assemblies (Dwg. 9332611) prior to loading. Each fuze assembly shall be radiographed and its radiographic image or hard copy radiograph examined before loading of the munition. (See 4.4.2.4)

3.4 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous, however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection (See 4.3).

b. Quality Conformance Inspection (See 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Pellet, Booster	8833581	10
Pad, Booster	9211053	5
Incendiary Briquet	9331307	10
Cover Male Hemisphere	9331889	10
Grenade, GP, HE, XM74 (Less fuze and explosives)	9332580	25
Male Hemisphere Assy.	9332613	4
Female Hemisphere Assy.	9332614	4
Male Hemisphere Loading Assy.	9332615	10
Female Hemisphere Loading Assy.	9332616	10
Cover Female Hemisphere	9332619	10

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-63393(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Grenade, GP, HE, XM74				See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Pellet, Booster (Dwg. 8833581)				
	Examination for defects	10	-	3.2	4.4.2.1
	Pad, Booster (Dwg. 9211053)				
	Examination for defects	5	-	3.2	4.4.2.3
	Incendiary Briquet (Dwg. 9331307)				
	Examination for defects	10	-	3.2	4.4.2.2
	Cover Male Hemisphere (Dwg. 9331889)				
	Conformance to the drawing	10	-	3.2	4.4.3.3
	Grenade, GP, HE, XM74 (less fuze and explosives (Dwg. 9332580)				
	Examination for defects (excluding 106)	15	-	3.2	4.4.2.10
	Pull test of crimp	10	-	3.2	4.5.3
	Male Hemisphere Assy. (Dwg. 9332613)				
	Examination for defects	4	-	3.2	4.4.2.6

NOTES:

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-63393(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Grenade, GP, HE, XM74				See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Female Hemisphere Assy. (Dwg. 9332614) Examination for defects	4	-	3.2	4.4.2.5
	Male Hemisphere Loading Assy. (Dwg. 9332615) Examination for defects	10	-	3.2	4.4.2.7
	Female Hemisphere Loading Assy. (Dwg. 9332616) Examination for defects	10	-	3.2	4.4.2.8
	Cover Female Hemisphere (Dwg. 9332619) Conformance to the drawing	10	-	3.2	4.4.3.3

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-63393(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Booster Pellet				8833581
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weight of booster pellet below min		0.40%	3.2	Balance
102	Height of booster pellet oversize		0.40%	3.2	Gage
103	Diameter of booster pellet oversize		0.40%	3.2	Gage
104	Booster pellet cracked		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-G-63393(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Incendiary Briquet		9331307
			NEXT HIGHER ASSEMBLY
			9332616 and 9332615

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Briquet cracked		0.40%	3.2	Visual
Minor 201	Mass of briquet incorrect		0.65%	3.2	Balance
202	Evidence of poor workmanship		1.00%	3.5	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-63393(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Pad, Booster			9211053
				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Diameter incorrect		0.40%	3.2	Gage
Minor 201 202	Thickness incorrect Evidence of poor workmanship		0.65% 1.00%	3.2 3.5	Gage Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-G-63393(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	M219A1E1 Grenade Fuze Assemblies Prior to Loading, Radiographic Examination				1933261l
					NEXT HIGHER ASSEMBLY —
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Fuze armed (See 6.4.1)		100%	3.3	4.5.4/x-ray
2	Fuze partially armed (See 6.4.2)		100%	3.3	4.5.4/x-ray
3	Fuze parts deformed		100%	3.3	4.5.4/x-ray
4	Fuze parts missing		100%	3.3	4.5.4/x-ray
5	Fuze parts out of position		100%	3.3	4.5.4/x-ray
6	Fuze parts redundant		100%	3.3	4.5.4/x-ray
Major	None defined				
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-G-63393(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Female Hemisphere Assy.		9332614
			NEXT HIGHER ASSEMBLY 9332616

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of inside diameter of cover flange extension to inside diameter of hemisphere interlock incorrect		0.40%	3.2	Gage
102	Depth from surface of cover flange to top of hemisphere interlock incorrect		0.40%	3.2	Gage
103	Inside diameter of hemisphere interlock undersize		0.40%	3.2	Gage
104	Torque test	(a)		3.2	4.5.1
Minor					
201	Weld on flute unsatisfactory		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

(a) See 4.4.3.1

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CLASSIFICATION OF DEFECTS & TESTS

MIL-G-63393(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Male Hemisphere Assy.		9332613
			NEXT HIGHER ASSEMBLY 9332615

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Depth from surface of cover flange to surface of top of hemisphere interlock		0.40%	3.2	Gage
102	Location of outside diameter of cover flange extension to outside diameter of hemisphere interlock incorrect		0.40%	3.2	Gage
103	Outside diameter of hemisphere interlock oversize		0.40%	3.2	Gage
104	Torque test	(a)		3.2	4.5.1
Minor					
201	Weld on flute unsatisfactory		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:
(a) See 4.4.3.1

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CLASSIFICATION OF DEFECTS & TESTS

MIL-G-63393(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.7	Male Hemisphere Loading Assy.		9332615
			NEXT HIGHER ASSEMBLY 9332580

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Depth from top of interlock to surface of charge oversize		0.40%	3.2	Gage
102	Charge above flush		0.40%	3.2	Gage
103	Depth from base of lower fuze cavity to base of booster cavity incorrect		0.40%	3.2	Gage
104	Depth from top of interlock to base of lower fuze cavity incorrect		0.40%	3.2	Gage
105	Depth from base of upper fuze cavity to base of lower fuze cavity incorrect		0.40%	3.2	Gage
106	Location of upper fuze cavity diameter to outside diameter of interlock incorrect		0.40%	3.2	Gage
107	Location of lower fuze cavity diameter incorrect		0.40%	3.2	Gage
108	Location of booster cavity diameter to upper fuze cavity diameter incorrect		0.40%	3.2	Gage
109	Diameter, upper fuze cavity, undersize		0.40%	3.2	Gage
110	Diameter, lower fuze cavity, undersize		0.40%	3.2	Gage
111	Diameter, booster pellet cavity, undersize		0.40%	3.2	Gage
112	Excessive voids and cavitation (Note 4)		0.40%	3.2	Gage
113	Density of cast load	(a)		3.2	4.5.2

NOTES: (a) See 4.4.3.2

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CLASSIFICATION OF DEFECTS & TESTS

MIL-G-63393(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.7	Male Hemisphere Loading Assy.				9332615
					NEXT HIGHER ASSEMBLY
					9332580
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Upper fuze cavity diameter, oversize		0.65%	3.2	Gage
202	Lower fuze cavity diameter, oversize		0.65%	3.2	Gage
203	Booster cavity diameter, oversize		0.65%	3.2	Gage
204	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Female Hemisphere Loading Assy.				9332616
					NEXT HIGHER ASSEMBLY 9332580
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Depth from base of interlock to surface of charge, oversize		0.40%	3.2	Gage
102	Charge above flush		0.40%	3.2	Gage
103	Depth from base of interlock to base of fuze cavity, incorrect		0.40%	3.2	Gage
104	Location of the fuze cavity diameter to inside diameter of interlock, incorrect		0.40%	3.2	Gage
105	Diameter, fuze cavity, undersize		0.40%	3.2	Gage
106	Excessive voids and cavitation (Note 4)		0.40%	3.2	Gage
107	Location of indentation in base of fuze cavity to fuze cavity diameter, incorrect		0.40%	3.2	Gage
108	Depth, indentation in base of fuze cavity, undersize		0.40%	3.2	Gage
109	Density of cast load	(a)		3.2	4.5.2
Minor					
201	Fuze cavity diameter, oversize		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES: (a) See 4.4.3.2

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Grenade, GP, HE XM74 Prior to Joining Male and Female Loading Assy.		9332580
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Booster pellet missing		0.40%	3.2	Visual
102	Booster pad missing		0.40%	3.2	Visual
103	Fuze missing		0.40%	3.2	Visual
104	Fuze inverted		0.40%	3.2	Visual
105	More than one booster pad		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-G-63393(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Grenade, GP, HE XM74				9332580
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Crimp missing or incorrect		0.40%	3.3	Visual
102	Cover torn, broken, or cracked		0.40%	3.3	Visual
103	Cover parted or deformed		0.40%	3.3	Visual
104	Flute not aligned		0.40%	3.3	Visual
105	Gap between mating surface of flanges exceeds max.		0.40%	3.2	Gage
106	Pull test of crimp	4.4.3.4		3.2	4.5.3
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual
grenade	Characteristics are to be checked during the first article inspection with the (less fuze and explosives).				

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4.4.3 Testing.

4.4.3.1 Torque test of male and female hemisphere assembly (Dwg. 9332613, 9332614). A sample of at least three (3) male or female hemisphere assemblies shall be tested in order to qualify the machine in the welding process for use in production. In addition, a random sample of three (3) male or female hemisphere assemblies shall be selected from the last portion of each four hours production for testing as a control of the tool during production. The torque test shall be performed as specified in 4.5.1.

4.4.3.2 Density of charge. Four female hemisphere loading assemblies and four male hemisphere loading assemblies shall be selected during each four hours production and their density measured as specified in 4.3.2. If any sample fails to conform to the applicable drawing requirement, that production quantity loaded since the last acceptable unit shall be rejected and returned to the contractor for corrective action.

4.4.3.3 Dimensional control of stamped parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts shall be dimensionally inspected for all drawing features to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hours production for dimensional inspection for all drawing features as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer.

4.4.3.4 Pull test of crimp (Dwg. 9332580). A sample of three (3) grenade assemblies, less fuze and explosives, shall be tested to qualify the machine in the crimping process for use in production.

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In addition, a sample of three (3) grenade assemblies, less fuze and explosives shall be selected from the last portion of each four (4) hours production for testing as a control of the tool during production. The pull test of crimp shall be performed as specified in 4.5.3.

If any grenade fails to comply with the pull test of crimp requirement specified on the applicable drawing, the grenade shall be classed defective and the crimping machine which produced the defective grenade shall be removed from production. Further, that portion of production since the last pull test inspection shall be returned to the contractor for corrective action. All crimp machines removed from production because of some fault, may, after reworking, be returned to production providing the grenades pass the pull test.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed in this specification is identified, either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the First Article or Quality Conformance Inspection Tables herein. The contractor shall submit inspection equipment designs to the Government for approval in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and 6.3 herein.

4.5 Test method and procedure.

4.5.1 Torque test of male and female hemisphere assembly. The hemisphere assembly shall be retained in the test fixture and the minimum required torque shall be applied at the specified rate. Any hemisphere assembly showing signs of loosening or deformation shall be classed defective and the welding machine which produced the defective part shall be removed from production. Further, that portion of production since the last torque test inspection should be returned to the contractor for corrective action. All welding machines removed from production because of fault, may, after reworking, be returned to production, providing the grenades pass the torque test (Destructive test).

4.5.2 Density of charge. The representative sample metal part hemisphere shall be weighed in air (without filler) and then weighed in water (without filler). The hemisphere is then loaded with filler under conditions identical in all respects with those employed in loading regular hemisphere. The hemisphere shall be drilled and then weighed in air, and in water. The density will be calculated as follows:

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$$\text{Density} = \frac{Wi - We}{(Wi - We) - (WWi - WWe)}$$

Where: We = Weight of empty hemisphere part in air
Wi = Weight of loaded and drilled assembly in air
WWe = Weight of empty hemisphere part in water
WWi = Weight of loaded and drilled assembly, in water

4.5.3 Pull test of crimp. Male and female hemisphere assemblies from the metal parts lot or lots being loaded shall be prepared by drilling a 5/8" hole in the center of each hemisphere assembly. An adapter suitable for mounting a bolt shall be placed in each hemisphere. A male and female hemisphere assembly shall then be joined and crimped in a manner identical in all respects with the employed items. Bolts shall be attached to the adapters through the holes in the hemisphere assemblies. Samples shall be tested in a tensile machine. The minimum specified load shall be applied at a rate not to exceed 125 pounds per second. The item shall be pull tested to destruction. All data generated shall be recorded.

4.5.4 X-Ray examination of M219A1E1 fuze assembly. The inspection equipment and the image quality of the radiographs for this examination shall be satisfactory to the procurement contracting officer. The direction of the X-ray beam employed in this inspection shall be parallel to the axis of spin of the fuze.

5. PACKAGING. Not applicable.

6. NOTES

6.1 Intended use. This Grenade is to be used in the Warhead M251E1.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to: Commander, US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND, ATTN: DRDAR-QAT-I, Dover, New Jersey 07801.

6.4 Definitions of critical defects for use in X-ray examination of fuze assemblies.

6.4.1 Armed fuze (critical defect). A fuze is defined as armed when the detonator rotor has turned approximately 90° from its normal unarmed position. The image of the detonator and the lead cup will be super-imposed on the radiograph (See 6.5).

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6.4.2 Partially armed fuze (critical defect). A fuze is defined as partially armed if: a) there is discernable rotation of the detonator rotor as evidenced by examination of the angle formed by the detonator image, the center of the fuze, and the lead cup image or b) there is no discernable rotation of the detonator rotor, however, two (2) or more adjacent lockweights are tipped away from the detonator rotor.

6.5 Disposition of defective fuze assemblies.

6.5.1 Critically defective fuze assemblies. When a critically defective fuze is found the procurement contracting officer shall be immediately notified. The fuze will be held in bonded storage pending disposition and in no instance will be destroyed without permission of the procurement contracting officer.

6.5.2 Fuzes with parts deformed, missing, out of position or redundant. Fuzes with parts deformed, missing, out of position, or redundant shall be destroyed in accordance with local safety requirements.

6.6 Specific instructions relative to manufacture, handling, storage and shipment of material:

6.6.1 All fuzes shall be inspected prior to release for loading in accordance with the following minimum procedures:

6.6.2 Orient individual fuzes (optional).

6.6.3 Mark fuze tray for identification.

6.6.4 Randomly introduce one inert, simulated-armed or partially-armed fuze per 2,000 fuzes in a manner that will preclude the possibility of the inspectors obtaining prior knowledge of the number and position of the simulated defects.

6.6.5 Record the position and tray number of each simulated defective introduced.

6.6.6 Staple or secure fuze packaging as required.

6.6.7 X-ray and inspect x-ray films or real time image.

6.6.8 Open as required those fuze packages containing defective fuzes, and replace rejected fuzes with good fuzes from inspected refill package of the same lot.

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6.6.9 Identify rejected fuzes by package number and position number within the package.

6.6.10 Restaple or resecure fuze packages as required.

6.6.11 As soon as possible after inspection, confirm that all simulated defective fuzes have been identified by the inspectors and removed from the lot. Should a simulated defective go undetected, the images inspected so far that day shall be re-inspected until all the undetected simulated units are identified. NOTE: When the inspection system identifies images to the operator who inspects them, it is only necessary to re-inspect those images reviewed by that operator during the shift in which the simulated defective was not detected.

6.6.12 As soon as possible after inspection, x-ray fuzes rejected for a critical defect to confirm that the correct fuze has been removed. CAUTION: Handle all fuzes suspected as having a critical defect as an armed fuze in accordance with local operating procedures.

6.6.13 While not obligatory, it is strongly recommended that all fuze packages from which reject fuzes have been removed for any reason and substituted for by refill fuzes be x-rayed again to confirm that all rejects have indeed been replaced by acceptable fuzes.

6.6.14 Mark fuze packages as having been x-rayed and inspected.

6.6.15 Fuze packages shall remain secured until fuzes reach point of assembly with end item. Prevent mixing with fuzes from other lots or with those that have not been x-ray inspected.

6.6.16 Following x-ray inspection, individual bare fuzes which are dropped any distance, to any surface, shall immediately be set aside, placed in properly identified containers and properly disposed of. Mishandled trays or boxes which are dropped any distance are to have all fuzes subsequently x-rayed and inspected. Fuzes spilled from packages, trays, or boxes shall be treated as armed fuzes. CAUTION: Any fuzes suspected of having been mishandled should be packed up only with tongs or other approved devices and not with the bare hands.

6.7 Distribution of ammunition data cards and test data. Distribution of data cards and functioning test data shall include the following: Commander, U.S. Army Armament Research and Development Command, DRDAR-QAT-M, Dover, New Jersey 07801

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6.8 Drawings. Drawings listed in Section 2 of this specification under the heading "US Army Armament Research and Development Command (ARRADCOM)" may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:
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Preparing activity:
Army-AR

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DOCUMENT IDENTIFIER (Number) AND TITLE MIL-G-63393(AR): Grenade, GP, HE, XM74. Loading, Assembling and Packing.

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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